MATH0006 Algebra 2

Year:	2026–2027
Code:	MATH0006
Level:	4 (UG)
Normal student group(s):	UG: Y1 Mathematics programmes
Value:	15 credits (7.5 ECTS credits)
Term:	2
Assessment:	85% examination, 15% coursework. In order to pass the module you must have at least 40% for both the examination mark and the final weighted mark.
Normal Pre-requisites:	MATH0005 (Algebra 1)
Lecturer:	Prof E Segal

Course Description and Objectives

This module will introduce students to the subject of abstract algebra, in which we define a class of algebraic structures by a system of axioms and then study their properties. We will begin with two examples which motivated the creation of the subject: modular arithmetic, and permutations. We will then meet our first axiomatic structure, groups. Groups capture the abstract idea of symmetry, they are a fundamental concept in algebra, and appear in most branches of mathematics. We will learn how to work with the group axioms, recognize examples (and non examples) of groups, and then move on to more advanced ideas such as subgroups and Lagrange's Theorem.

The next structures we study will be rings, which generalize the integers, and fields, which generalize the rational or real numbers. We will see how many properties of the integers can be generalized to other rings, in particular learning the theory behind factorizing polynomials. In the final section we introduce vector spaces, which are the abstract setting for the vectors and matrices studied in Algebra 1.

Recommended Texts

You are not required to buy any books for this course, but you may find the following helpful:

– M Liebeck, *A Concise Introduction to Pure Mathematics*

– C Whitehead, *A Guide to Abstract Algebra*

– CR Jordan and DA Jordan, *Groups*

Detailed Syllabus

Basic number theory:

– Division theorem, Euclidean Algorithm, Bezout's Lemma. Prime factorisation.

– Infinitely many primes.

– Modular arithmetic. Invertibility mod n.

Permutations:

– Permutations. Cycles. Notation. Cycle type. Order.

- Sign of a permutation. Multiplicativity of sign (proof non-examinable).

Groups:

- Axioms of a group. Examples ($\mathbb{Z}$,$\mathbb{Z}_n$,S_n,GL_n,D_n,$\mathbb{R}^n$,...). Non-examples.
- Abelian groups. Cayley tables.
- Deducing first properties from axioms: uniqueness of inverse/identity, inverse of an inverse and product, cancellation.
- Order of an element. Cyclic groups.
- Conjugation. Cycle types of permutations as conjugacy classes.
- Isomorphisms.
- Direct products.
- Subgroups. Examples. Subgroups of cyclic groups.
- Cosets. Lagrange's Theorem. Corollaries.
- Fermat's Little Theorem

Rings:

- Definition of a ring. Examples ($\mathbb{Z}$,$\mathbb{Z}_n$, Gaussian integers...).
- Units, zero divisors, integral domains.
- Irreducible elements.
- Fields. $\mathbb{Z}_p$ as a field. Characteristic.
- Polynomial rings. Degree, division and remainder.
- Unique factorization for polynomials.

Vector spaces:

- Definition and examples, ($\mathbb{R}^n$,$\mathbb{F}^n$, polynomials, function spaces, ..).
- Subspaces. A subspace is a vector space.
- Spanning, linear independence, bases (generalized from Algebra 1).
- Dimension. Finite-dimensionality. Examples of infinite-dimensional spaces.
- A finite field has order p^k.
- Linear maps. Examples (matrices, differentiation and integration, geometric examples on $\mathbb{R}^2$,...).
- Non-examples.
- Expressing a linear map as a matrix.
- Kernels, image, rank-nullity (generalized from Algebra 1).